

Time-dependent variables and multiple states

The following practical exercise is based on the data from paper:

P Hovind, L Tarnow, P Rossing, B Carstensen, and HH Parving: Improved survival in patients obtaining remission of nephrotic range albuminuria in diabetic nephropathy. *Kidney Int*, **66**(3):1180–1186, Sept 2004.

You can find a .pdf-version of the paper here: <http://BendixCarstensen.com/AdvCoh/papers/Hovind.2004.pdf>

The renal failure dataset

The dataset `renal.dta` contains data on follow up of 125 patients from Steno Diabetes Center. They enter the study when they are diagnosed with nephrotic range albuminuria (NRA). This is a condition where the levels of albumin in the urine is exceeds a certain level as a sign of kidney disease. The levels may however drop as a consequence of treatment, this is called remission. Patients exit the study at death or kidney failure (dialysis or transplant).

Variable	Description
id	Patient id
sex	1=male, 2=female
dob	Date of birth
doe	Date of entry into the study (2.5 years after NRA)
dor	Date of remission. Missing if no remission has occurred
dox	Date of exit from study
event	Exit status: 1,2,3=event (death, ESRD), 0=censored

1. The dataset is in Stata-format, so you must read the dataset using `read.dta` from the `foreign` package (which is part of the standard R-distribution). At the same time, convert `sex` to a proper factor. Choose where to read the dataset.

```
library(Epi)
library(survival)
library(mgcv, quietly = TRUE)
```

This is mgcv 1.9-4. For overview type '?mgcv'.

```
library(tidyverse, quietly = TRUE)
```

```
-- Attaching core tidyverse packages -----
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2

-- Conflicts -----
x dplyr::collapse() masks nlme::collapse()
x dplyr::filter()   masks stats::filter()
x dplyr::lag()      masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(foreign)
# renal <- read.dta(
#   "https://raw.githubusercontent.com/SPE-R/SPE/master/pracs-book/data/renal.dta")
renal <- read.dta("http://BendixCarstensen.com/SPE/data/renal.dta")
```

```
renal$sex <- factor(renal$sex, labels = c("M", "F"))
head(renal)
```

2. Use the Lexis function to declare the data as survival data with age, calendar time and time since entry into the study as timescales. Label any event > 0 as *ESRD*, i.e. renal death (death of kidney (transplant or dialysis), or person). Note that you must make sure that the *alive* state (here *NRA*) is the first, as Lexis assumes that everyone starts in this state (unless of course `entry.status` is specified):

```
Lr <- Lexis(entry = list(per = doe,
                        age = doe - dob,
                        tfi = 0),
            exit = list(per = dox),
            exit.status = factor(event > 0, labels = c("NRA", "ESRD")),
            data = renal)
```

NOTE: `entry.status` has been set to "NRA" for all.

```
summary(Lr)
```

Transitions:

To

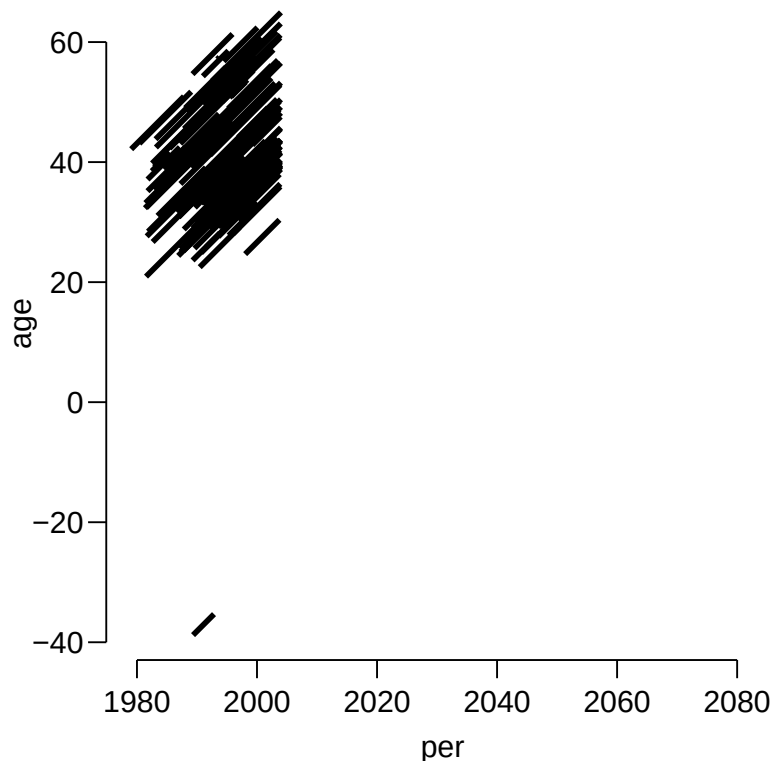
From NRA ESRD Records: Events: Risk time: Persons:

NRA	48	77	125	77	1084.67	125
-----	----	----	-----	----	---------	-----

Make sure you know what the variables in `Lr` stand for.

3. Visualize the follow-up in a Lexis-diagram, by using the `plot` method for Lexis objects.

```
plot(Lr, col = "black", lwd = 3)
```



```
subset(Lr, age < 0)
```

```
lex.id    per    age tfi lex.dur lex.Cst lex.Xst id sex    dob    doe dor    dox
      88 1989.34 -38.81  0    3.5    NRA    ESRD 586  M 2028.15 1989.34  NA 1992.84
event
      1
```

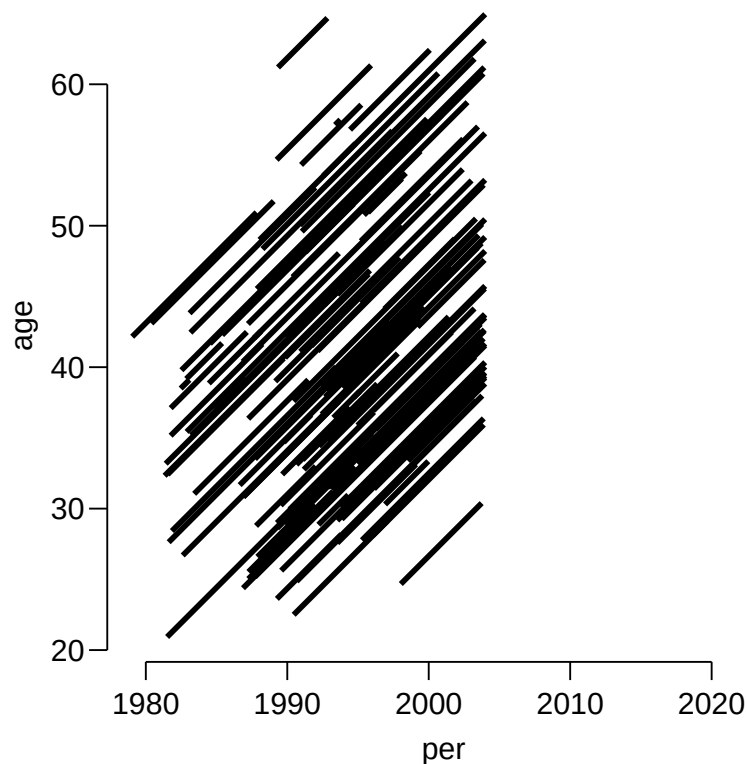
What is wrong here? List the data for the person with negative entry age.

4. Correct the data and make a new plot, for example by:

```
Lr <- transform(Lr, dob = ifelse(dob > 2000, dob - 100, dob),
                             age = ifelse(dob > 2000, age + 100, age))
subset(Lr, id == 586)
```

```
lex.id    per    age tfi lex.dur lex.Cst lex.Xst id sex    dob    doe dor    dox
      88 1989.34 61.19  0    3.5    NRA    ESRD 586  M 1928.15 1989.34  NA 1992.84
event
      1
```

```
plot(Lr, col = "black", lwd = 3)
```



5. Now make a Cox-regression analysis of ESRD occurrence with the variables sex and age at entry into the study, using time since entry to the study as time scale.

```
mc <- coxph(Surv(lex.dur, lex.Xst == "ESRD")
            ~ I(age / 10) + sex, data = Lr)
summary(mc)
```

Call:

```
coxph(formula = Surv(lex.dur, lex.Xst == "ESRD") ~ I(age/10) +
```

```
sex, data = Lr)

n= 125, number of events= 77

      coef exp(coef) se(coef)      z Pr(>|z|)
I(age/10) 0.5514    1.7357   0.1402  3.932 8.43e-05
sexF      -0.1817    0.8338   0.2727 -0.666  0.505

      exp(coef) exp(-coef) lower .95 upper .95
I(age/10)    1.7357    0.5761    1.3186    2.285
sexF         0.8338    1.1993    0.4886    1.423

Concordance= 0.612 (se = 0.036 )
Likelihood ratio test= 16.07 on 2 df,  p=3e-04
Wald test              = 16.38 on 2 df,  p=3e-04
Score (logrank) test = 16.77 on 2 df,  p=2e-04
```

What is the hazard ratio between males and females? Between two persons who differ 10 years in age at entry?

6. The main focus of the paper was to assess whether the occurrence of remission (return to a lower level of albumin excretion, an indication of kidney recovery) influences mortality. *Remission* is a time-dependent variable which is initially 0, but takes the value 1 when remission occurs. In order to handle this, each person who sees a remission must have two records:
 - One record for the time before remission, where entry is `doe`, exit is `dor`, remission is 0, and event is 0.
 - One record for the time after remission, where entry is `dor`, exit is `dox`, remission is 1, and event is 0 or 1 according to whether the person had an event at `dox`.

This is accomplished using the `cutLexis` function on the `Lexis` object, where we introduce a remission state `Rem`. Also use `split.state=TRUE` to have different ESRD states according to whether a person had had remission or not prior to ESRD. The statement to do this is:

```
Lc <- cutLexis(Lr, cut = Lr$dor, # where to cut follow up
               timescale = "per", # what timescale are we referring to
               new.state = "Rem", # name of the new state
               split.state = TRUE) # different states depending on previous
summary(Lc)
```

Transitions:

	To							
From	NRA	Rem	ESRD	ESRD(Rem)	Records:	Events:	Risk time:	Persons:
NRA	24	29	69	0	122	98	824.77	122
Rem	0	24	0	8	32	8	259.90	32
Sum	24	53	69	8	154	106	1084.67	125

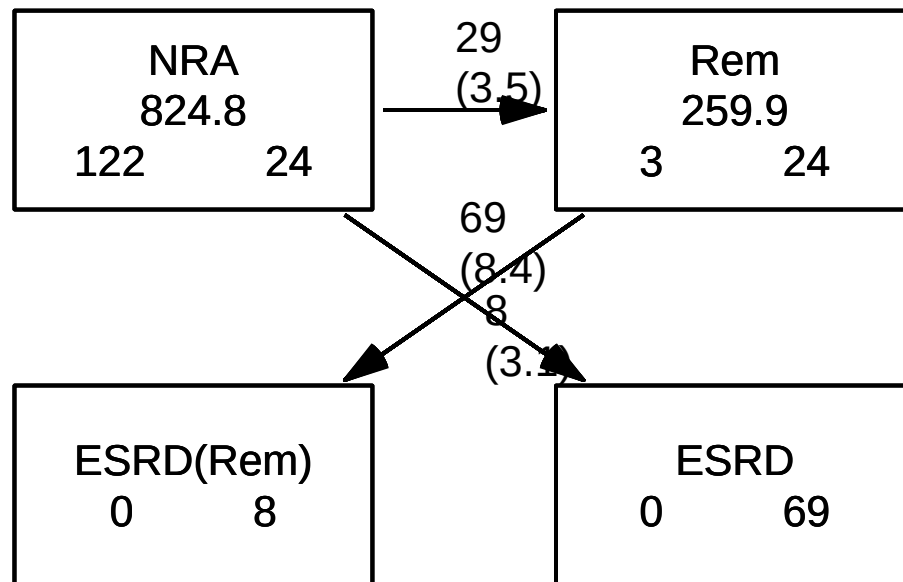
List the records from a few select persons (choose values for `lex.id`, using for example `subset(Lc, lex.id %in% c(5,7,9))`).

7. Now show how the states are connected and the number of transitions between them by using `boxes`. This is an interactive command that requires you to click in the graph window:

```
boxes(Lc)
```

It has a couple of fancy arguments, try:

```
boxes(Lc, boxpos = TRUE, scale.R = 100, show.BE = TRUE, hm = 1.5, wm = 1.5)
```

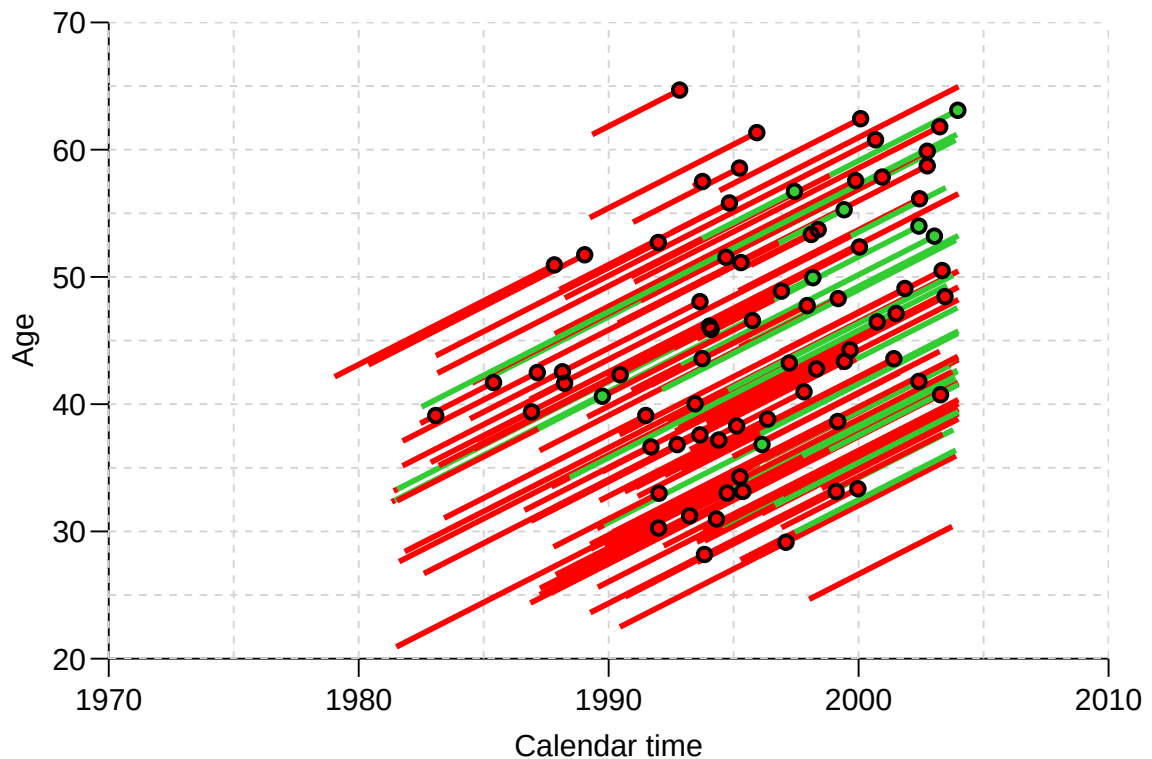


That is presumably not what you want, so you may be tempted to read the help page for `boxes.Lexis` ...

8. Draw a Lexis diagram where different coloring is used for different segments of the follow-up. The `plot.Lexis` function draws a line for each record in the dataset, so you can index the coloring by `lex.Cst` and `lex.Xst` as appropriate — indexing by a factor corresponds to indexing by the *index number* of the factor levels, so you must be know which order the factor levels are in:

```
levels(Lc) # names and order of states in lex.Cst and lex.Xst
```

```
[1] "NRA"      "Rem"      "ESRD"     "ESRD(Rem)"
par(mai = c(3, 3, 1, 1) / 4, mgp = c(3, 1, 0) / 1.6)
plot(Lc, col = c("red", "limegreen")[Lc$lex.Cst],
     xlab = "Calendar time", ylab = "Age",
     lwd = 3, grid = 0:20 * 5, las = 1,
     xlim = c(1970, 2010), ylim = c(20, 70),
     xaxs = "i", yaxs = "i")
points(Lc, pch = c(NA, NA, 16, 16)[Lc$lex.Xst],
       col = c("red", "limegreen", "transparent", "transparent")[Lc$lex.Cst])
points(Lc, pch = c(NA, NA, 1, 1)[Lc$lex.Xst],
       col = "black", lwd = 2)
```



9. Make a Cox-regression of mortality rates (i.e. endpoint ESRD or ESRD(Rem)) with sex, age at entry and remission as explanatory variables, using time since entry as timescale, and include `lex.Cst` as time-dependent variable, and indicate that each record represents follow-up from `tfi` to `tfi+lex.dur`. Make sure that you know why what goes where in the call to `coxph`.

```
(EP <- levels(Lc)[3:4]) # define EndPoint states
```

```
[1] "ESRD"      "ESRD(Rem)"
```

```
m1 <- coxph(Surv(tfi,           # entry time
                 tfi + lex.dur, # exit time
                 lex.Xst %in% EP) # event
             ~ sex + I((doe - dob - 40) / 10) + # fixed covariates
               (lex.Cst == "Rem"),             # time-dependent variable
             data = Lc)
summary(m1)
```

Call:

```
coxph(formula = Surv(tfi, tfi + lex.dur, lex.Xst %in% EP) ~ sex +
      I((doe - dob - 40)/10) + (lex.Cst == "Rem"), data = Lc)
```

n= 154, number of events= 77

	coef	exp(coef)	se(coef)	z	Pr(> z)
sexF	-0.05534	0.94616	0.27500	-0.201	0.840517
I((doe - dob - 40)/10)	0.52190	1.68522	0.13655	3.822	0.000132
lex.Cst == "Rem"TRUE	-1.26241	0.28297	0.38483	-3.280	0.001036

	exp(coef)	exp(-coef)	lower .95	upper .95
sexF	0.94616	1.05698	0.59881	1.51808
I((doe - dob - 40)/10)	1.68522	0.59371	1.28881	2.10811
lex.Cst == "Rem"TRUE	0.28297	3.53611	0.14881	6.88111

sexF	0.9462	1.0569	0.5519	1.6220
I((doe - dob - 40)/10)	1.6852	0.5934	1.2895	2.2024
lex.Cst == "Rem"TRUE	0.2830	3.5339	0.1331	0.6016

```

Concordance= 0.664 (se = 0.033 )
Likelihood ratio test= 30.31 on 3 df, p=1e-06
Wald test = 27.07 on 3 df, p=6e-06
Score (logrank) test = 29.41 on 3 df, p=2e-06

```

```
ci.exp(m1)
```

	exp(Est.)	2.5%	97.5%
sexF	0.9461646	0.5519334	1.621985
I((doe - dob - 40)/10)	1.6852196	1.2895097	2.202360
lex.Cst == "Rem"TRUE	0.2829710	0.1330996	0.601599

There is actually also a wrapper for a Cox model used in a Lexis object—since the baseline timescale has a special status this is given as the l.h.s. of the formula:

```

mL <- coxphLexis(Lc, tfi ~ sex + I((doe - dob - 40) / 10) + # fixed covariates
                                (lex.Cst == "Rem"))          # time-dependent variable

```

survival::coxph analysis of Lexis object Lc:

Rates for transitions:

NRA->ESRD

Rem->ESRD(Rem)

Baseline timescale: tfi

```
ci.exp(mL)
```

	exp(Est.)	2.5%	97.5%
sexF	0.9461646	0.5519334	1.621985
I((doe - dob - 40)/10)	1.6852196	1.2895097	2.202360
lex.Cst == "Rem"TRUE	0.2829710	0.1330996	0.601599

What is the effect of remission on the rate of ESRD?

Splitting the follow-up time

In order to explore the effect of remission on the rate of ESRD, we split the data further into small pieces of follow-up. To this end we again use the function `splitLexis`. The rates can then be modeled using a Poisson-model, and the shape of the effect of the underlying *rates* be explored. Furthermore, we can allow effects of both time since NRA and current age. To this end we will use splines, so we need the `splines` and also the `mgcv` packages.

- Now split the follow-up time every month after entry, and verify that the number of events and risk time is the same as before and after the split:

```

sLc <- splitLexis(Lc, "tfi", breaks = seq(0, 30, 1/12))
summary(sLc)

```

Transitions:

To

From	NRA	Rem	ESRD	ESRD(Rem)	Records	Events	Risk time	Persons
NRA	24	29	69	0	122	98	824.77	122
Rem	0	24	0	8	32	8	259.90	32
Sum	24	53	69	8	154	106	1084.67	125

```
summary(sLc)
```

Transitions:

```
      To
From  NRA  Rem ESRD ESRD(Rem) Records: Events: Risk time: Persons:
  NRA 9854   29   69         0    9952     98    824.77      122
  Rem   0 3139   0         8    3147     8    259.90       32
  Sum 9854 3168  69         8   13099    106   1084.67     125
```

What is the difference between the datasets Lc and sLc?

11. Now fit the Poisson-model corresponding to the Cox-model we fitted previously. The function `Ns()` produces a model matrix corresponding to a piece-wise cubic function, modeling the baseline hazard explicitly (think of the `Ns` terms as the baseline hazard that is not visible in the Cox-model). You can use the wrapper function `glmLexis`

```
mp <- glmLexis(sLc,
  ~ Ns(tfi, knots = c(0, 2, 5, 10)) +
    sex + I((doe - dob - 40) / 10) +
    I(lex.Cst == "Rem"))
```

stats::glm Poisson analysis of Lexis object sLc with log link:

Rates for transitions:

NRA->ESRD

Rem->ESRD(Rem)

```
ci.exp(mp)
```

	exp(Est.)	2.5%	97.5%
(Intercept)	0.01664432	0.003956666	0.07001685
Ns(tfi, knots = c(0, 2, 5, 10))1	5.18917655	1.949197027	13.81469029
Ns(tfi, knots = c(0, 2, 5, 10))2	34.20004199	1.764818735	662.75524463
Ns(tfi, knots = c(0, 2, 5, 10))3	4.43318269	2.179977108	9.01528219
sexF	0.91751162	0.536258443	1.56981691
I((doe - dob - 40)/10)	1.70082390	1.300813859	2.22384004
I(lex.Cst == "Rem")TRUE	0.27927558	0.131396852	0.59358233

How does the effects of sex change from the Cox-model?

12. Try instead using the `gam` function from the `mgcv` package. There is convenience wrapper for this for Lexis objects as well:

```
mx <- gamLexis(sLc,
  ~ s(tfi, k = 10) +
    sex + I((doe - dob - 40) / 10) +
    I(lex.Cst == "Rem"))
```

mgcv::gam Poisson analysis of Lexis object sLc with log link:

Rates for transitions:

NRA->ESRD

Rem->ESRD(Rem)

```
ci.exp(mp, subset = c("Cst", "doe", "sex"))
```

	exp(Est.)	2.5%	97.5%
I(lex.Cst == "Rem")TRUE	0.2792756	0.1313969	0.5935823
I((doe - dob - 40)/10)	1.7008239	1.3008139	2.2238400
sexF	0.9175116	0.5362584	1.5698169


```
ci.exp(mx, subset = c("Cst", "doe", "sex"))
```

	exp(Est.)	2.5%	97.5%
I(lex.Cst == "Rem")TRUE	0.2784659	0.1309446	0.5921838
I((doe - dob - 40)/10)	1.6992069	1.2995225	2.2218192
sexF	0.9309945	0.5435486	1.5946150

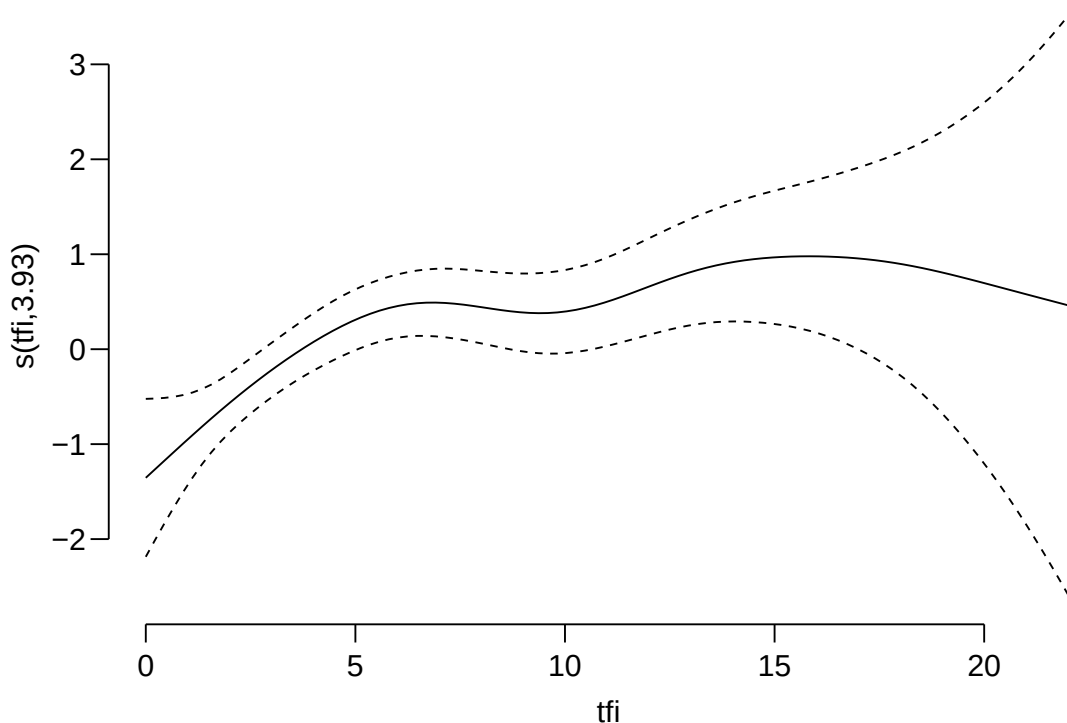
```
ci.exp(m1, subset = c("Cst", "doe", "sex")) # from Cox model
```

	exp(Est.)	2.5%	97.5%
lex.Cst == "Rem"TRUE	0.2829710	0.1330996	0.601599
I((doe - dob - 40)/10)	1.6852196	1.2895097	2.202360
sexF	0.9461646	0.5519334	1.621985

We see that there is virtually no difference between the three approaches in terms of the regression parameters.

- The model has the same assumptions as the Cox-model about proportionality of rates, but there is an additional assumption that the hazard is a smooth function of time since entry. It seems to be a sensible assumption (well, restriction) to put on the rates that they vary smoothly by time. No such restriction is made in the Cox model. The `gam` model optimizes the shape of the smoother by general cross-validation. Try to look at the shape of the estimated effect of `tfi`:

```
plot(mx)
```



Is this a useful plot?

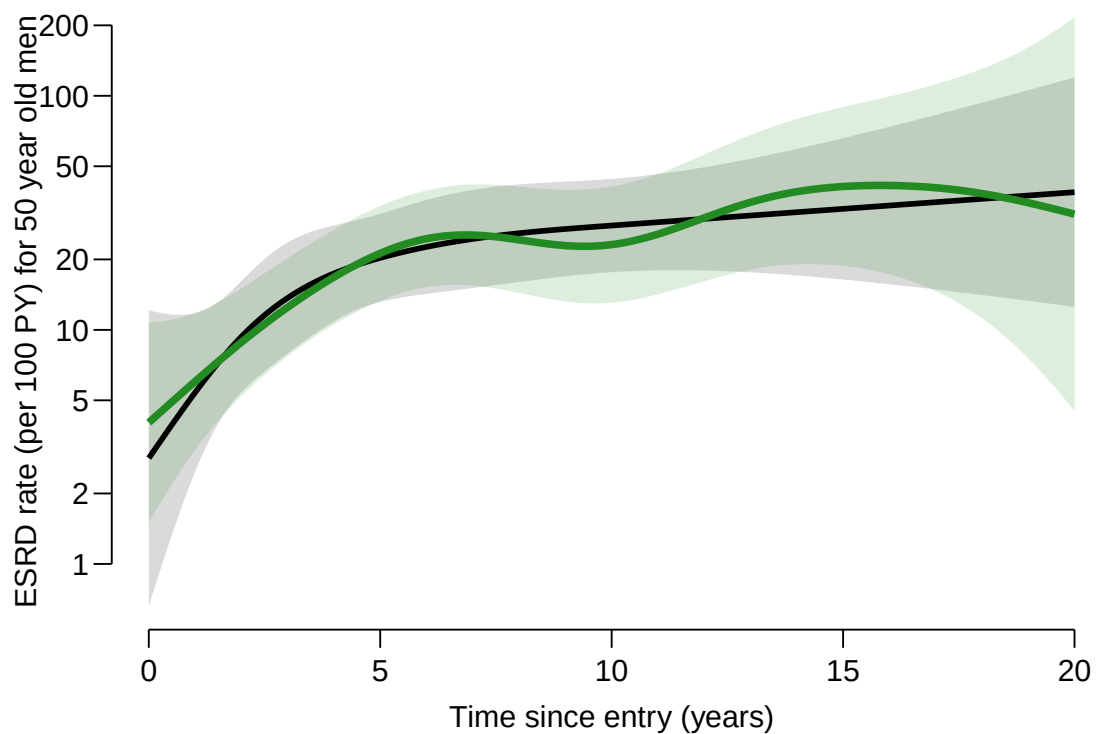
- However, `plot` (well, `plot.gam`) does not give you the *absolute* level of the underlying rates because it bypasses the intercept. So in order to predict the rates as a function of `tfi` and the covariates, we set up a prediction data frame. Note that age in the model specification is entered as `doe-dob`, hence the prediction data frame must have these two variables and not the age, but it is only the difference that

matters for the prediction:

```
nd <- data.frame(tfi = seq(0, 20, 0.1),
                 sex = "M",
                 doe = 1990,
                 dob = 1940,
                 lex.Cst = "NRA")
str(nd)

'data.frame': 201 obs. of 5 variables:
 $ tfi      : num  0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 ...
 $ sex      : chr  "M" "M" "M" "M" ...
 $ doe      : num  1990 1990 1990 1990 1990 1990 1990 1990 1990 1990 ...
 $ dob      : num  1940 1940 1940 1940 1940 1940 1940 1940 1940 1940 ...
 $ lex.Cst  : chr  "NRA" "NRA" "NRA" "NRA" ...

matshade(nd$tfi, cbind(ci.pred(mp, newdata = nd),
                      ci.pred(mx, newdata = nd)) * 100,
          plot = TRUE,
          type = "l", lwd = 3:4, col = c("black", "forestgreen"),
          log = "y", xlab = "Time since entry (years)",
          ylab = "ESRD rate (per 100 PY) for 50 year old men")
```



Try to overlay with the corresponding prediction from the glm model using `Ns`. Any preferences?

Prediction from the multistate model

If we want to make proper statements about the survival and disease probabilities we must know not only how the occurrence of remission influences the rate of death/ESRD, but we must also model the occurrence rate of remission itself.

17. The rates of ESRD were modelled by a Poisson model with effects of age and time since NRA — in the models `mp` and `mx`. But if we want to model whole process we must also model the remission rates transition from NRA to Rem, but the number of events is rather small so we restrict covariates in this model to only time since NRA and sex. Note that only the records that represent follow-up in the NRA state should be used; this is most easily done using the `gamLexis` function

```
mr <- gamLexis(sLc, ~ s(tfi, k = 10) + sex,
               from = "NRA",
               to = "Rem")
```

```
mgcv::gam Poisson analysis of Lexis object sLc with log link:
```

```
Rates for the transition:
```

```
NRA->Rem
```

```
summary(mr)
```

```
Family: poisson
```

```
Link function: log
```

```
Formula:
```

```
cbind(trt(Lx$lex.Cst, Lx$lex.Xst) %in% trnam, Lx$lex.dur) ~ s(tfi,
  k = 10) + sex
```

```
Parametric coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.7025	0.2582	-14.339	<2e-16
sexF	0.9579	0.3728	2.569	0.0102

```
Approximate significance of smooth terms:
```

	edf	Ref.df	Chi.sq	p-value
s(tfi)	1.013	1.025	0.066	0.813

```
R-sq.(adj) = -5.65e-06 Deviance explained = 1.65%
```

```
UBRE = -0.96024 Scale est. = 1 n = 9952
```

```
ci.exp(mr, pval = TRUE)
```

	exp(Est.)	2.5%	97.5%	P
(Intercept)	0.02466174	0.01486718	0.04090901	1.254019e-46
sexF	2.60620470	1.25503844	5.41202779	1.019130e-02
s(tfi).1	1.00499489	0.89131271	1.13317662	9.351638e-01
s(tfi).2	0.99623769	0.80778743	1.22865188	9.718940e-01
s(tfi).3	0.99822247	0.91911268	1.08414140	9.663137e-01
s(tfi).4	1.00188999	0.89006998	1.12775801	9.750528e-01
s(tfi).5	0.99842904	0.92280785	1.08024715	9.687920e-01
s(tfi).6	0.99817367	0.90142303	1.10530865	9.719666e-01
s(tfi).7	1.00168704	0.92615949	1.08337380	9.663850e-01
s(tfi).8	0.99448155	0.68445450	1.44493689	9.768400e-01
s(tfi).9	0.94790052	0.63349476	1.41834700	7.946918e-01

What is the remission rate-ratio between men and women?

18. If we want to predict the probability of being in each of the three states using these estimated rates, we may resort to analytical calculations of the probabilities from the estimated rates, which is actually doable in this case, but which will be largely intractable for more complicated models. Alternatively we can *simulate* the life course for a large group of (identical) individuals through a model using the

estimated rates. That will give a simulated cohort (in the form of a `Lexis` object), and we can then just count the number of persons in each state at each of a set of time points. This is accomplished using the function `simLexis`. The input to this is the initial status of the persons whose life-course we shall simulate, and the transition rates in suitable form:

- Suppose we want predictions for men aged 50 at NRA. The input is in the form of a `Lexis` object (where `lex.dur` and `lex.Xst` will be ignored). Note that in order to carry over the `time.scales` and the `time.since` attributes, we construct the input object using `subset` to select columns, and `NULL` to select rows (see the example in the help file for `simLexis`):

```
inL <- subset(sLc, select = 1:11)[NULL, ]
str(inL)
```

Classes 'Lexis' and 'data.frame': 0 obs. of 11 variables:

```
$ lex.id : int
$ per    : num
$ age    : num
$ tfi    : num
$ lex.dur: num
$ lex.Cst: Factor w/ 4 levels "NRA","Rem","ESRD",...
$ lex.Xst: Factor w/ 4 levels "NRA","Rem","ESRD",...
$ id     : num
$ sex    : Factor w/ 2 levels "M","F":
$ dob    : num
$ doe    : num
- attr(*, "time.scales")= chr [1:3] "per" "age" "tfi"
- attr(*, "time.since")= chr [1:3] "" "" ""
- attr(*, "breaks")=List of 3
..$ per: NULL
..$ age: NULL
..$ tfi: num [1:361] 0 0.0833 0.1667 0.25 0.3333 ...
```

```
timeScales(inL)
```

```
[1] "per" "age" "tfi"
```

```
inL[1, "lex.id"] <- 1
inL[1, "per"] <- 2000
inL[1, "age"] <- 50
inL[1, "tfi"] <- 0
inL[1, "lex.Cst"] <- "NRA"
inL[1, "lex.Xst"] <- NA
inL[1, "lex.dur"] <- NA
inL[1, "sex"] <- "M"
inL[1, "dob"] <- 1950
inL[1, "doe"] <- 2000
inL <- rbind(inL, inL)
inL[2, "sex"] <- "F"
inL
```

```
lex.id per age tfi lex.dur lex.Cst lex.Xst id sex dob doe
1 2000 50 0 NA NRA <NA> NA M 1950 2000
1 2000 50 0 NA NRA <NA> NA F 1950 2000
```

```
str(inL)
```

Classes 'Lexis' and 'data.frame': 2 obs. of 11 variables:

```
$ lex.id : num 1 1
```

```

$ per      : num  2000 2000
$ age      : num   50  50
$ tfi      : num    0   0
$ lex.dur: num  NA NA
$ lex.Cst: Factor w/ 4 levels "NRA","Rem","ESRD",...: 1 1
$ lex.Xst: Factor w/ 4 levels "NRA","Rem","ESRD",...: NA NA
$ id       : num  NA NA
$ sex      : Factor w/ 2 levels "M","F": 1 2
$ dob      : num  1950 1950
$ doe      : num  2000 2000
- attr(*, "breaks")=List of 3
..$ per: NULL
..$ age: NULL
..$ tfi: num [1:361] 0 0.0833 0.1667 0.25 0.3333 ...
- attr(*, "time.scales")= chr [1:3] "per" "age" "tfi"
- attr(*, "time.since")= chr [1:3] "" "" ""

```

The other input for the simulation is the models for the possible transitions.

This is given as a list with an element for each transient state (that is **NRA** and **Rem**), each of which is again a list with names equal to the states that can be reached from the transient state. The content of the list will be **glm** objects, in this case the models we just fitted, describing the transition rates:

```

Tr <- list("NRA" = list("Rem" = mr,
                        "ESRD" = mx),
          "Rem" = list("ESRD(Rem)" = mx))

```

With this as input we can now generate a cohort, using **N=5** to simulate life course of 10 persons (5 for each set of starting values in **inL**):

```

(iL <- simLexis(Tr, inL, N = 5))

```

lex.id	per	age	tfi	lex.dur	lex.Cst	lex.Xst	id	sex	dob	doe	cens
1	2000.00	50.00	0.00	4.31	NRA	ESRD	NA	M	1950	2000	2020
2	2000.00	50.00	0.00	4.82	NRA	ESRD	NA	M	1950	2000	2020
3	2000.00	50.00	0.00	16.17	NRA	ESRD	NA	M	1950	2000	2020
4	2000.00	50.00	0.00	7.00	NRA	Rem	NA	M	1950	2000	2020
4	2007.00	57.00	7.00	13.00	Rem	Rem	NA	M	1950	2000	2020
5	2000.00	50.00	0.00	5.26	NRA	ESRD	NA	M	1950	2000	2020
6	2000.00	50.00	0.00	14.11	NRA	ESRD	NA	F	1950	2000	2020
7	2000.00	50.00	0.00	13.29	NRA	ESRD	NA	F	1950	2000	2020
8	2000.00	50.00	0.00	12.67	NRA	ESRD	NA	F	1950	2000	2020
9	2000.00	50.00	0.00	5.30	NRA	Rem	NA	F	1950	2000	2020
9	2005.30	55.30	5.30	14.70	Rem	Rem	NA	F	1950	2000	2020
10	2000.00	50.00	0.00	7.09	NRA	Rem	NA	F	1950	2000	2020
10	2007.09	57.09	7.09	8.20	Rem	ESRD(Rem)	NA	F	1950	2000	2020

```

summary(iL, by = "sex")

```

\$M

Transitions:

	To								
From	NRA	Rem	ESRD	ESRD(Rem)	Records:	Events:	Risk time:	Persons:	
NRA	0	1	4	0	5	5	37.56	5	
Rem	0	1	0	0	1	0	13.00	1	
Sum	0	2	4	0	6	5	50.57	5	

\$F

Transitions:

	To							
From	NRA	Rem	ESRD	ESRD(Rem)	Records:	Events:	Risk time:	Persons:
NRA	0	2	3	0	5	5	52.46	5
Rem	0	1	0	1	2	1	22.90	2
Sum	0	3	3	1	7	6	75.36	5

What type of object have you got as `iL`?

19. Now generate the life course of, say, 1,000 persons, and look at the summary. The `system.time` command is just to tell you how long it took, you may want to start with 500 just to see how long that takes.

```
system.time(sM <- simLexis(Tr, inL, N = 5000, t.range = 12))
```

```
bruger    system forløbet
20.65     1.64    23.09
```

```
summary(sM, by = "sex")
```

\$M

Transitions:

	To							
From	NRA	Rem	ESRD	ESRD(Rem)	Records:	Events:	Risk time:	Persons:
NRA	400	685	3915	0	5000	4600	27640.7	5000
Rem	0	406	0	279	685	279	4529.3	685
Sum	400	1091	3915	279	5685	4879	32170.0	5000

\$F

Transitions:

	To							
From	NRA	Rem	ESRD	ESRD(Rem)	Records:	Events:	Risk time:	Persons:
NRA	275	1615	3110	0	5000	4725	23843.01	5000
Rem	0	957	0	658	1615	658	11159.40	1615
Sum	275	2572	3110	658	6615	5383	35002.41	5000

Why are there so many ESRD-events in the resulting data set?

20. Now count how many persons are present in each state at each time for the first 10 years after entry (which is at age 50). This can be done by using `nState`. Try:

```
nStm <- nState(subset(sM, sex == "M"), time.scale = "age",
               at = seq(0, 10, 0.1),
               from = 50)
nStf <- nState(subset(sM, sex == "F"), time.scale = "age",
               at = seq(0, 10, 0.1),
               from = 50)
head(nStf, 15)
```

	State			
when	NRA	Rem	ESRD	ESRD(Rem)
50	5000	0	0	0
50.1	4952	32	16	0

50.2	4881	77	42	0
50.3	4820	118	62	0
50.4	4752	158	90	0
50.5	4697	186	117	0
50.6	4643	224	133	0
50.7	4587	253	159	1
50.8	4528	286	185	1
50.9	4451	326	222	1
51	4393	355	250	2
51.1	4326	396	275	3
51.2	4267	427	303	3
51.3	4220	456	320	4
51.4	4161	486	349	4

What is in the object `nStf`?

21. With the counts of persons in each state at the designated time points (in `nStm`), compute the cumulative fraction over the states, arranged in order given by `perm`:

```
ppm <- pState(nStm, perm = c(2, 1, 3, 4))
ppf <- pState(nStf, perm = c(2, 1, 3, 4))
head(ppf)
```

	State			
when	Rem	NRA	ESRD	ESRD(Rem)
50	0.0000	1.0000	1	1
50.1	0.0064	0.9968	1	1
50.2	0.0154	0.9916	1	1
50.3	0.0236	0.9876	1	1
50.4	0.0316	0.9820	1	1
50.5	0.0372	0.9766	1	1

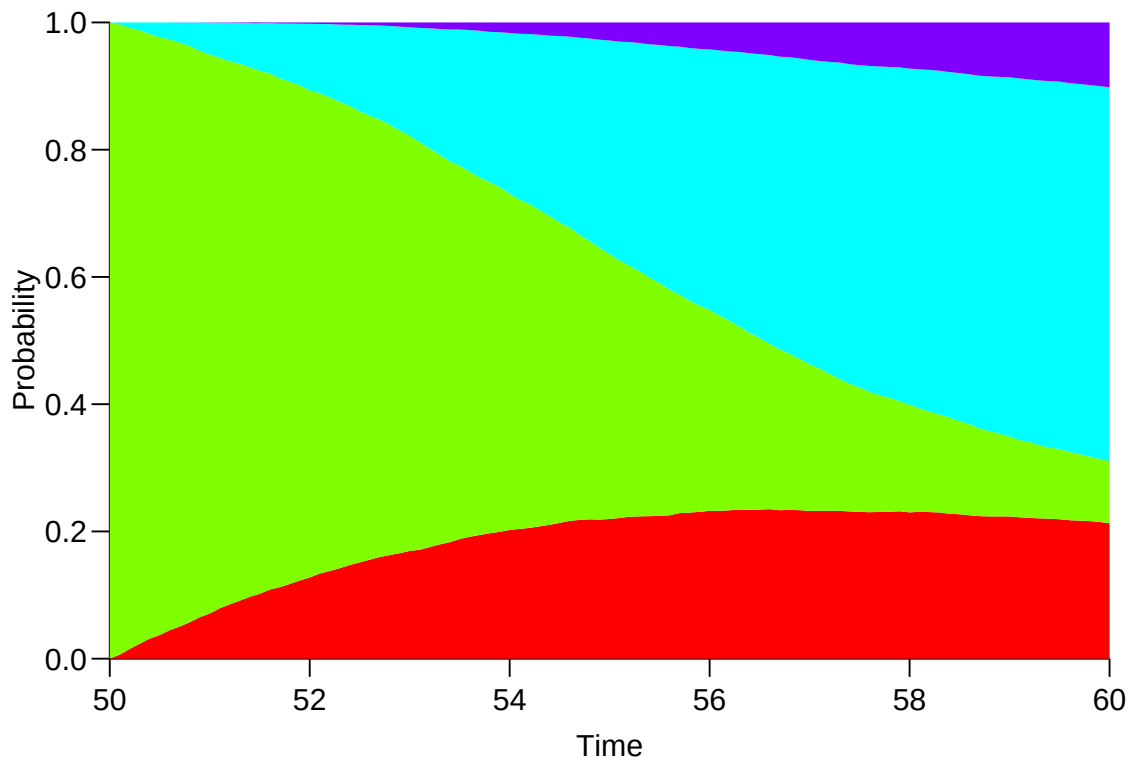
```
tail(ppf)
```

	State			
when	Rem	NRA	ESRD	ESRD(Rem)
59.5	0.2192	0.3294	0.9070	1
59.6	0.2176	0.3250	0.9046	1
59.7	0.2170	0.3216	0.9032	1
59.8	0.2162	0.3176	0.9016	1
59.9	0.2148	0.3140	0.9000	1
60	0.2130	0.3106	0.8980	1

What do the entries in `ppf` represent?

22. Try to plot the cumulative probabilities using the `plot` method for `pState` objects:

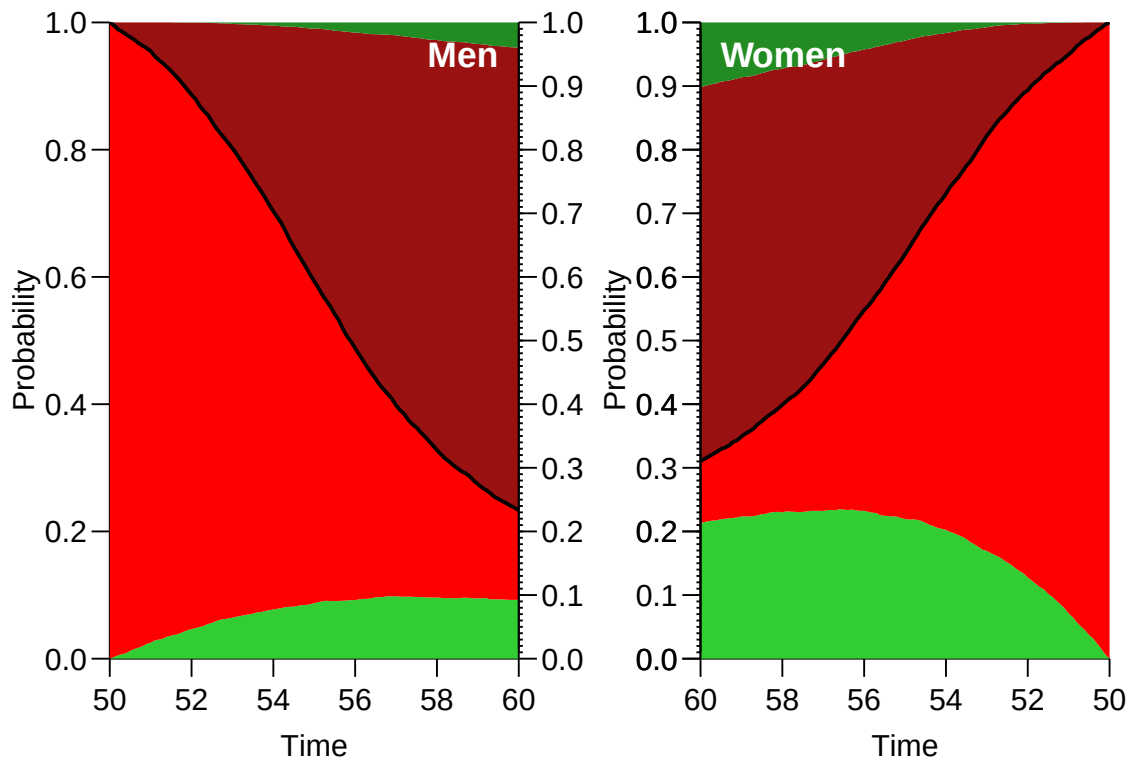
```
plot(ppf)
```



Is this useful?

23. Now try to improve the plot so that it is easier to read, and easier to compare between men and women, for example:

```
par(mfrow = c(1, 2))
# Men
plot(ppm, col = c("limegreen", "red", "#991111", "forestgreen"))
lines(as.numeric(rownames(ppm)), ppm[, "NRA"], lwd = 2)
text(59.5, 0.95, "Men", adj = 1, col = "white", font = 2, cex = 1.2)
axis(side = 4, at = 0:10 / 10)
axis(side = 4, at = 1:99 / 100, labels = NA, tck = -0.01)
# Women
plot(ppf, col = c("limegreen", "red", "#991111", "forestgreen"),
      xlim = c(60, 50)) # inverted x-axis
lines(as.numeric(rownames(ppf)), ppf[, "NRA"], lwd = 2)
text(59.5, 0.95, "Women", adj = 0, col = "white", font = 2, cex = 1.2)
axis(side = 2, at = 0:10 / 10)
axis(side = 2, at = 1:99 / 100, labels = NA, tck = -0.01)
```

What is the 10-year risk of remission for 50 year old men and women respectively?